Dear Editors,

I am pleased to submit the manuscript entitled “Folded Transport MCMC: Eliminating Label Switching by Sampling on a Fundamental Domain” for consideration for publication in the Journal of Computational and Graphical Statistics.

**Summary.** In Bayesian mixture models and other exchangeable-component models, the posterior is invariant under permutation of component labels, creating $m!$ equivalent modes—the label-switching problem. Standard practice runs MCMC on the full labelled posterior and relabels the samples afterwards, but this leaves the underlying chain to traverse all $m!$ symmetric modes and provides no guarantee that it has done so. The paper proposes Folded Transport MCMC (FolT-MCMC), which eliminates label switching *before* sampling by restricting the Markov chain to a fundamental domain—a sorted or reflected subspace containing one representative per symmetric mode.

The method has two computational ingredients: a learned normalising flow, symmetrised over the group orbits to give a valid proposal on the fundamental domain; and a computable convergence diagnostic based on the oscillation of the log-density ratio between target and proposal. The central finding is that label switching destroys this diagnostic—the cross-mode oscillation inflates it to a vacuous value even when the sampler performs well—and that folding restores it.

**Main contributions.** (1) A fundamental-domain sampler with an exact identity relating the folded and original credible sets, and a symmetrised quotient proposal. (2) A proof that the convergence diagnostic transfers to the fundamental domain with strictly improved inputs (smaller credible-set diameter, larger density floor, smaller covering radius), and is therefore sharper than the diagnostic on the original space whenever the unfolded flow under-represents the symmetric modes. (3) Empirical validation on Gaussian mixtures ($d$ = 2–20), label-switching targets (up to 24 equivalent modes), a standard Bayesian three-component mixture posterior, and real accelerometer data from a supertall building, with diagnostic improvement ratios of $2\times$ to $145\times$.

**Why JCGS.** The paper is a computational and methodological contribution to Bayesian computation: it combines a familiar idea from the mixture-model literature (sampling on a sorted chamber) with modern learned proposals and a practical convergence diagnostic, and demonstrates the result on both synthetic and real data. A standard Bayesian mixture benchmark shows that post-hoc relabelling can recover correct point estimates yet provide no convergence guarantee, clarifying when quotient-space sampling is actually needed. We believe the mix of method, theory, and reproducible computation fits the scope of JCGS well.

**Related work by the author.** The convergence diagnostic builds on two preprints by the same author (arXiv:2606.01078 and arXiv:2605.30722). The supplementary material restates the key results from these works in a self-contained appendix, so the present paper can be read and verified without reference to them.

**Declarations.** This manuscript has not been published previously and is not under consideration at any other journal. Code will be released upon acceptance. There are no conflicts of interest to declare.

Sincerely,

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